

Input



Peer Review
Text



Paper Text

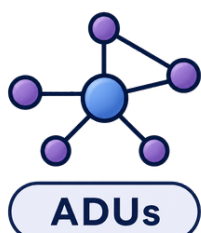
Preprocessing

Data Segmentation & Preprocessing Unit



Extract
Sections
Unit

- Summary
- Strengths
- Weaknesses
- Questions



LLM Extraction /
Argument
Discourse Units
Extraction Unit



LLM Extraction/
Atomic Review
Comments, Flaws,
Claims, Contributions

Modular Automated Evaluation Core (LLM Judge)

(1) Depth of Analysis



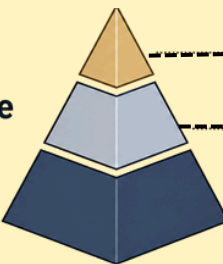
Classify Role
(Claim / Premise)

& Classify Aspect
(Novelty, Methodology,
Experiment, Clarity)

Is Role =
Premise?

LLM Judge chain
atomic Review
Comments (Yes)

Evidence Grounding



Level 0 - Vague
Level 1 - Internal
Grounding
Level 2 - External
Grounding

Evaluate
Depth

Targeted Literature Retrieval



(2) Novelty
Assessment



Build Queries
(ID = core task +
contribution)



Semantic
Scholar API



Candidate Pool
(ID + Title + Abstract)

Automated Review Verification



Pair-Wise
Matching
Claim-Candidate



LLM Judge
Classify Score
per Pair

Flaw Verification & Severity (LLM Judge)



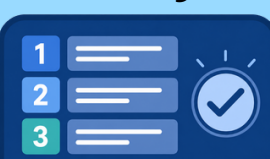
(3) Flaw
Identification &
Major Issues



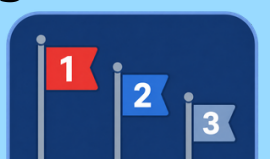
Is Flaw
Valid?



Critical / Minor / None

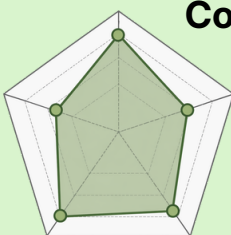


Align with
Review & Assess
Prioritization



Critique Order
Flags

(4) Multi-dimensional Constructiveness



Constructiveness
Profile

Constructiveness Dimension Scoring



Actionability



Specificity



Justification



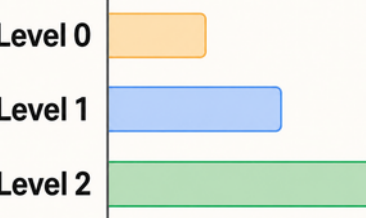
Solution
Orientation



Tone &
Professionalism

Metric Output

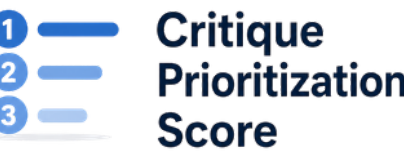
Depth of Analysis Score



Novelty Grounding Score



Flaw Identification Score



Constructiveness Score

